In [1]:
```
#Anabela Petrushevikj
#D207
#Exploratory Analysis
```

In [2]:
```
#Import all packages
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

In [3]:
```
df = pd.read_csv('medical_clean_D207.csv')
```

In [4]:
```
#Getting information about the dataset and it's variables
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 50 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   CaseOrder           10000 non-null  int64
 1   Customer_id         10000 non-null  object
 2   Interaction         10000 non-null  object
 3   UID                 10000 non-null  object
 4   City                10000 non-null  object
 5   State               10000 non-null  object
 6   County              10000 non-null  object
 7   Zip                 10000 non-null  int64
 8   Lat                 10000 non-null  float64
 9   Lng                 10000 non-null  float64
 10  Population          10000 non-null  int64
 11  Area                10000 non-null  object
 12  TimeZone            10000 non-null  object
 13  Job                 10000 non-null  object
 14  Children            10000 non-null  int64
 15  Age                 10000 non-null  int64
 16  Income              10000 non-null  float64
 17  Marital             10000 non-null  object
 18  Gender              10000 non-null  object
 19  ReAdmis             10000 non-null  object
 20  VitD_levels         10000 non-null  float64
 21  Doc_visits          10000 non-null  int64
 22  Full_meals_eaten    10000 non-null  int64
 23  vitD_supp           10000 non-null  int64
 24  Soft_drink          10000 non-null  object
 25  Initial_admin       10000 non-null  object
 26  HighBlood           10000 non-null  object
 27  Stroke              10000 non-null  object
 28  Complication_risk   10000 non-null  object
 29  Overweight          10000 non-null  object
 30  Arthritis           10000 non-null  object
 31  Diabetes            10000 non-null  object
 32  Hyperlipidemia      10000 non-null  object
 33  BackPain            10000 non-null  object
 34  Anxiety             10000 non-null  object
 35  Allergic_rhinitis   10000 non-null  object
 36  Reflux_esophagitis  10000 non-null  object
 37  Asthma              10000 non-null  object
 38  Services            10000 non-null  object
 39  Initial_days        10000 non-null  float64
 40  TotalCharge         10000 non-null  float64
 41  Additional_charges  10000 non-null  float64
 42  Item1               10000 non-null  int64
 43  Item2               10000 non-null  int64
 44  Item3               10000 non-null  int64
 45  Item4               10000 non-null  int64
 46  Item5               10000 non-null  int64
 47  Item6               10000 non-null  int64
 48  Item7               10000 non-null  int64
 49  Item8               10000 non-null  int64
dtypes: float64(7), int64(16), object(27)
memory usage: 3.8+ MB
```

In [5]:

```
# Separate the Initial_days data for patients who were readmitted and those who were not
initial_days_readmitted = df[df['ReAdmis'] == 'Yes']['Initial_days']
```

```
initial_days_not_readmitted = df[df['ReAdmis'] == 'No']['Initial_days']

# Perform an independent t-test
t_stat_initial_days, p_value = stats.ttest_ind(initial_days_readmitted, initial_days_not_readmit

print('T-test statistic:',t_stat_initial_days)
print('P-value:', p_value)

alpha = 0.05

if p_value < alpha:
    print('There is a significant relationship between readmission of patients and the initial nu
else:
    print('There is not a significant relationship between readmission of patients and the initia
```

```
T-test statistic: 203.00996600080393
P-value: 0.0
There is a significant relationship between readmission of patients and the initial number of day
s that patients were hospitalized.
```

In [6]:

```
#Univariate Statistics for Initial_days variable
summary_initial_days = df['Initial_days'].describe()
print('Summary for Initial_days:', summary_initial_days)
```

```
Summary for Initial_days: count    10000.000000
mean        34.455299
std         26.309341
min          1.001981
25%          7.896215
50%         35.836244
75%         61.161020
max         71.981490
Name: Initial_days, dtype: float64
```

In [7]:

```
#Univariate Statistics for Age variable
summary_age = df['Age'].describe()
print('Summary for Age:', summary_age)
```

```
Summary for Age: count    10000.000000
mean        53.511700
std         20.638538
min         18.000000
25%         36.000000
50%         53.000000
75%         71.000000
max         89.000000
Name: Age, dtype: float64
```

In [8]:

```
#Univariate Statistics for ReAdmis variable
readmission = df['ReAdmis'].value_counts(normalize = True).sort_values(ascending = False)
print('Relative frequency for Readmission:', readmission)
```

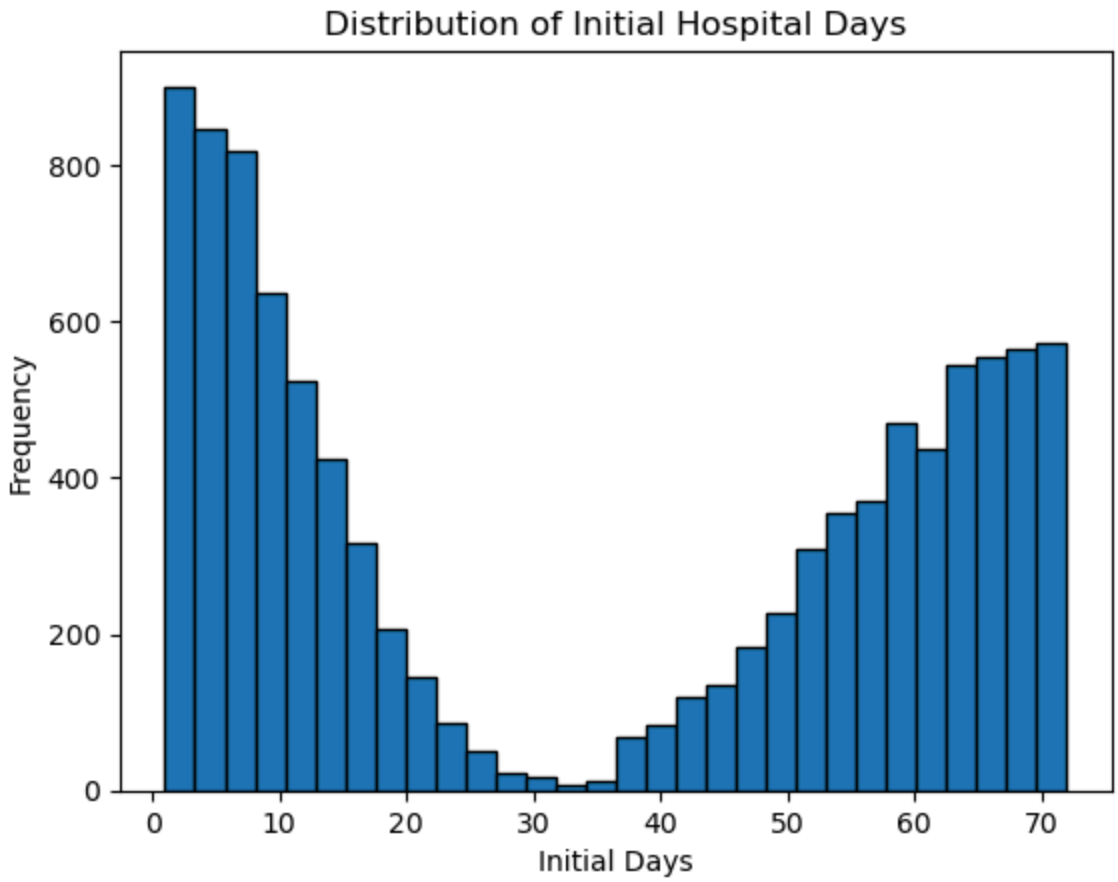
```
Relative frequency for Readmission: ReAdmis
No     0.6331
Yes    0.3669
Name: proportion, dtype: float64
```

In [9]:

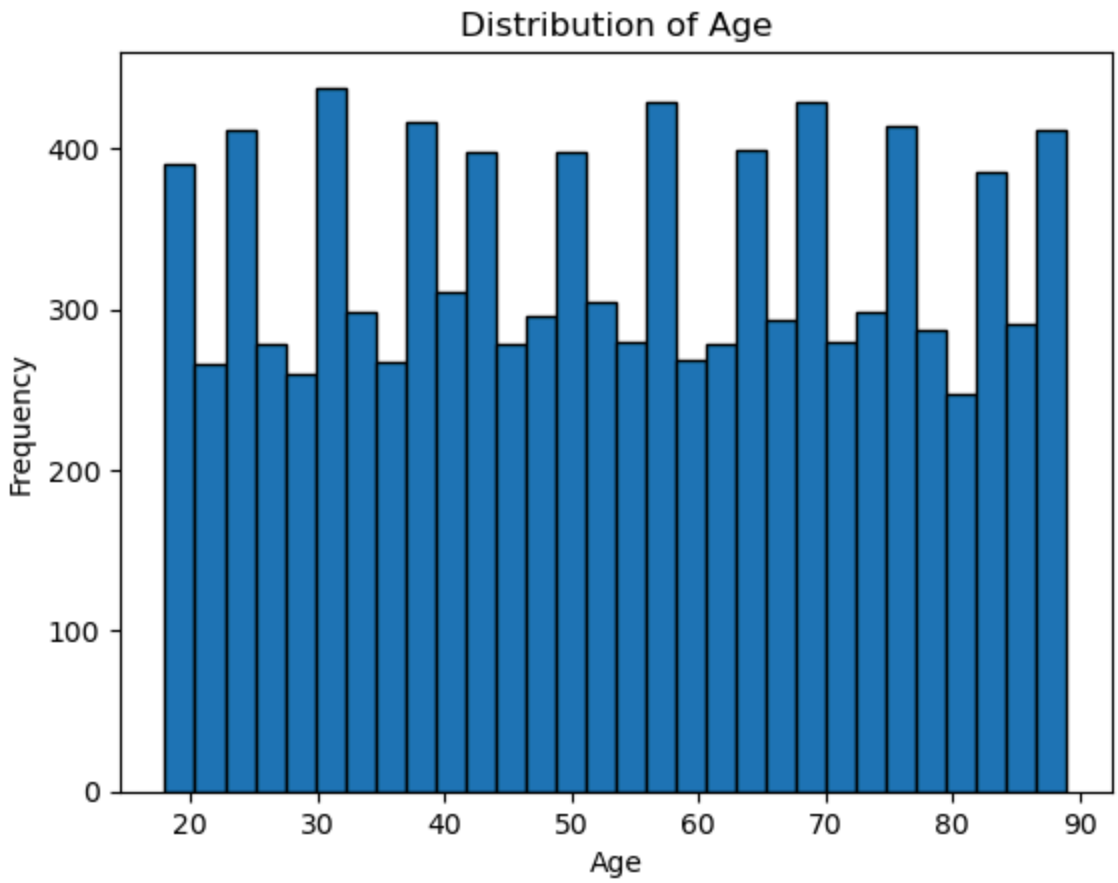
```
#Univariate Statistics for Complication_Risk variable
complication_risk = df['Complication_risk'].value_counts(normalize = True).sort_values(ascending
print('Relative frequency for complication risk:', complication_risk)
```

```
Relative frequency for complication risk: Complication_risk
Medium    0.4517
High      0.3358
Low       0.2125
Name: proportion, dtype: float64
```

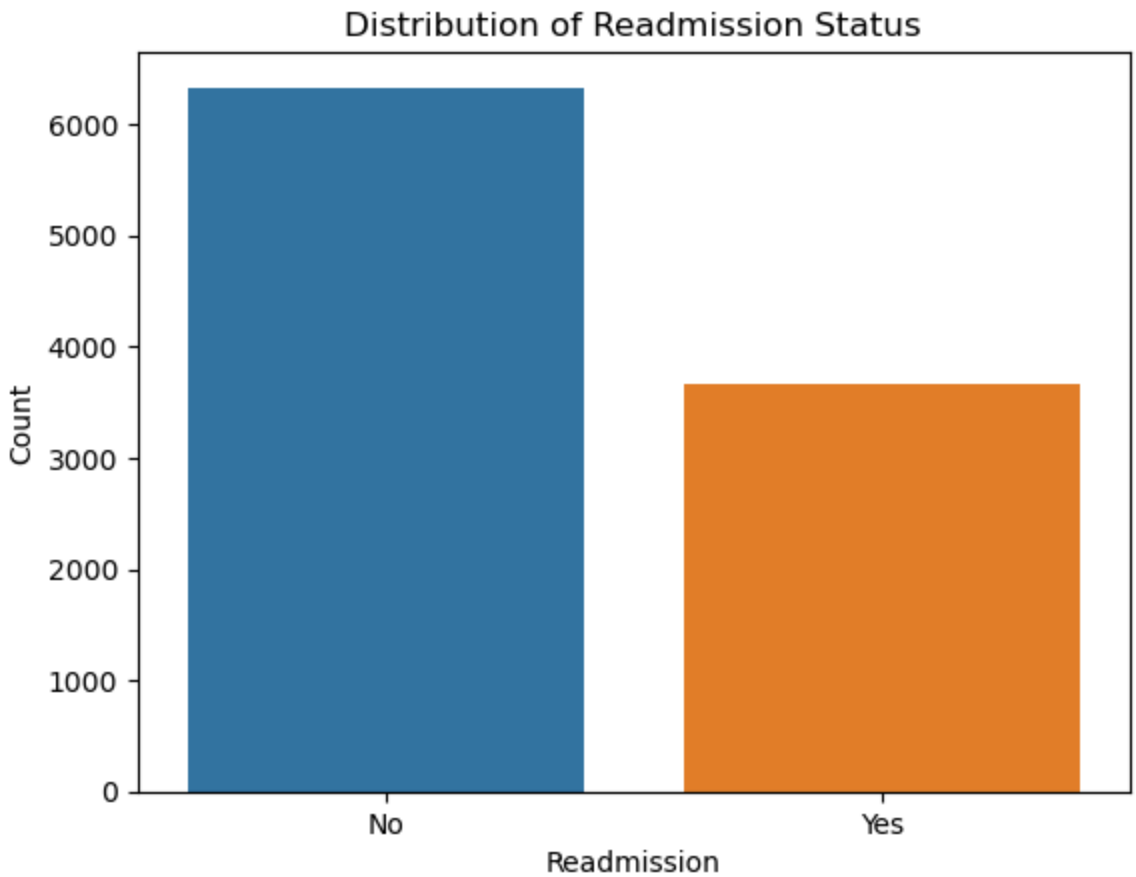
In [10]:
```python
# Continuous variable 1: Distribution of Initial_days
plt.hist(df['Initial_days'], bins=30, edgecolor='black')
plt.title('Distribution of Initial Hospital Days')
plt.xlabel('Initial Days')
plt.ylabel('Frequency')
plt.show()
```



In [11]:
```python
# Continuous variable 2: Distribution of Age
plt.hist(df['Age'], bins=30, edgecolor='black')
plt.title('Distribution of Age')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```

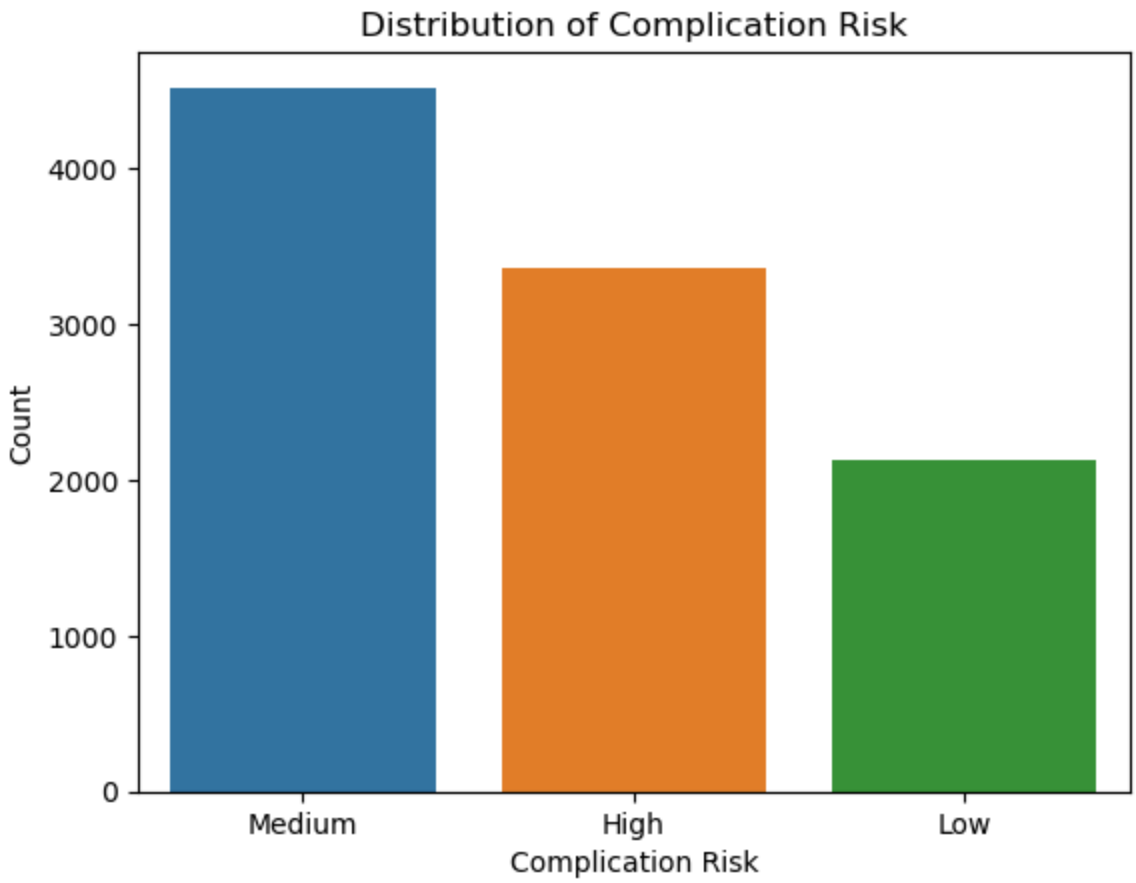


In [12]:
```
# Categorical variable 1: Distribution of ReAdmis
sns.countplot(x='ReAdmis', hue= None, data=df)
plt.title('Distribution of Readmission Status')
plt.xlabel('Readmission')
plt.ylabel('Count')
plt.show()
```



In [13]:
```
# Categorical variable 2: Distribution of Complication Risk
sns.countplot(x='Complication_risk', hue= None, data=df)
plt.title('Distribution of Complication Risk')
plt.xlabel('Complication Risk')
plt.ylabel('Count')
```

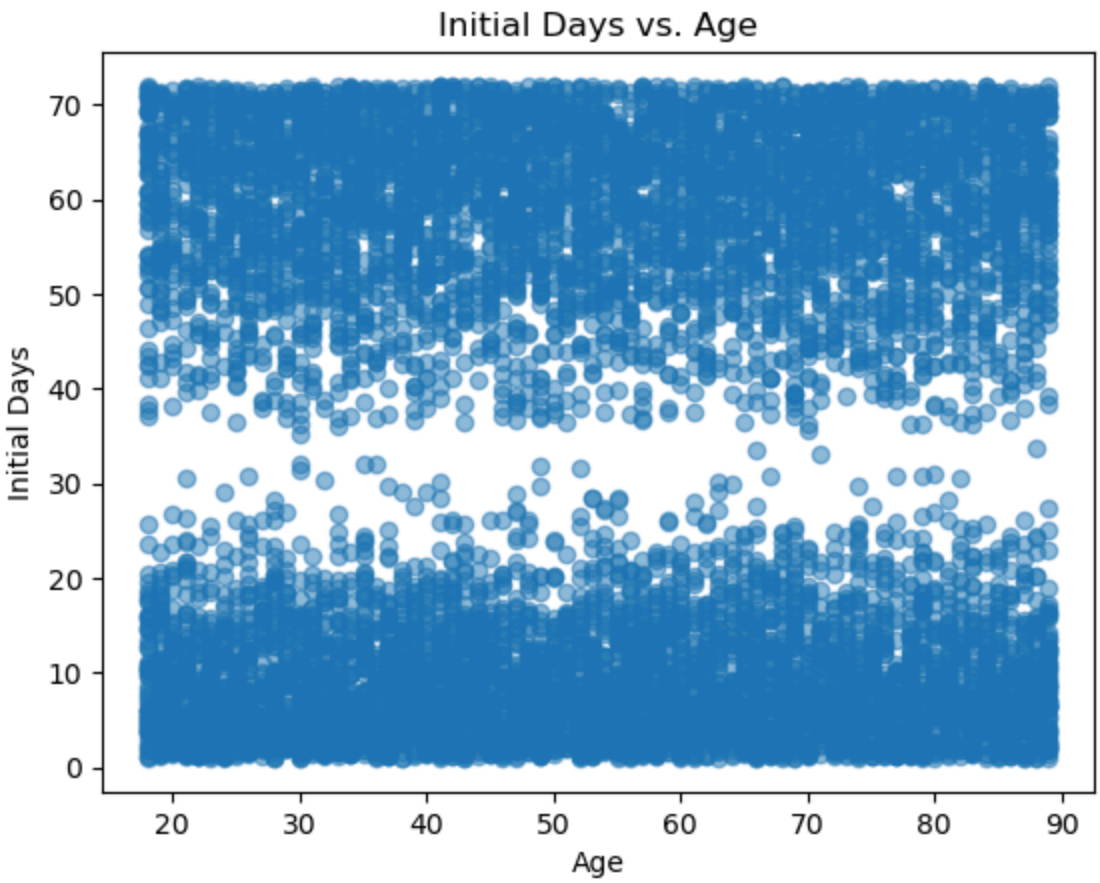
Out[13]:
```
Text(0, 0.5, 'Count')
```



In [14]:
```
#Bivariate analysis for Initial_days and Age variables
pearson, _ = stats.pearsonr(df.Initial_days, df.Age)
print("Pearson correlation coefficient:")
print(pearson)
```

```
Pearson correlation coefficient:
0.016264289866716027
```

In [15]:
```
# Bivariate plot 1: Scatter plot of Initial Days vs. Age
plt.scatter(df['Age'], df['Initial_days'], alpha=0.5)
plt.title('Initial Days vs. Age')
plt.xlabel('Age')
plt.ylabel('Initial Days')
plt.show()
```



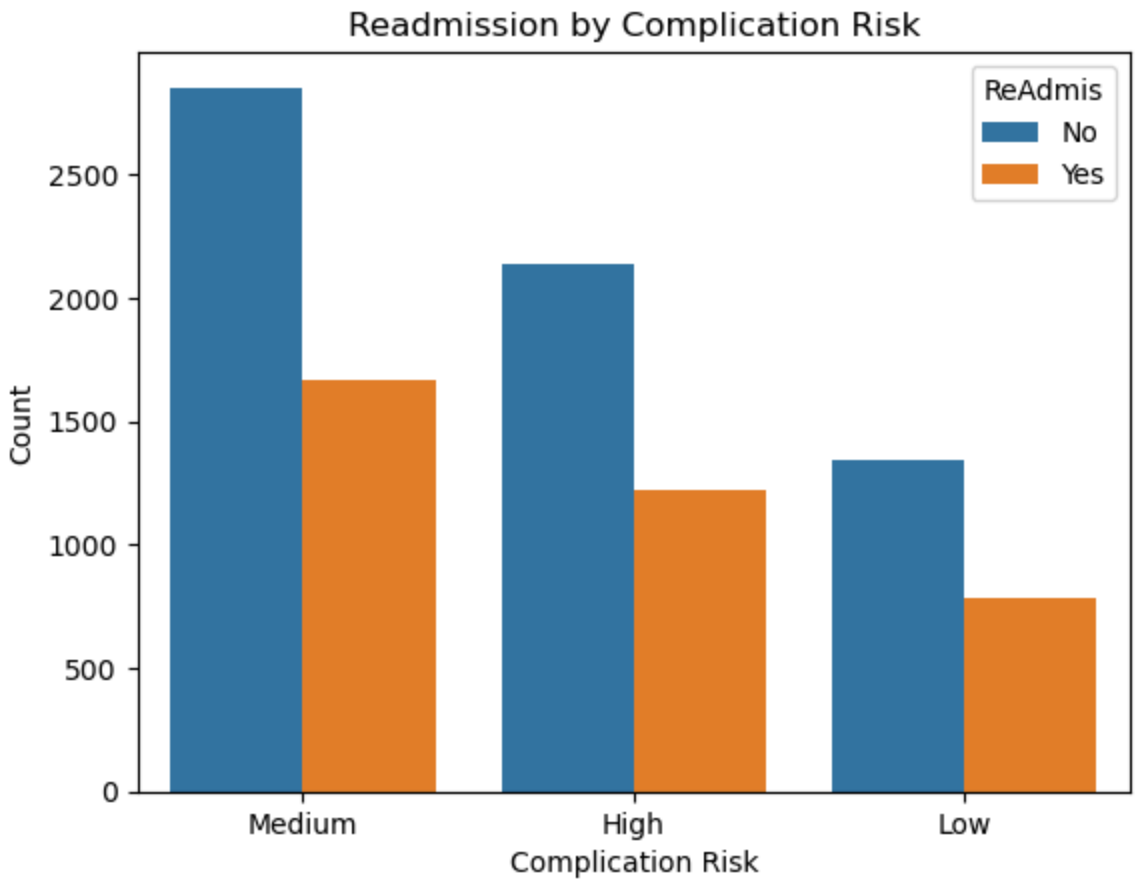
In [16]:
```
#Bivariate analysis for Readmission and Complication Risk
#Contingency table
contingency_table = pd.crosstab(df['ReAdmis'], df['Complication_risk'])

# Perform the chi-square test of independence
chi2, p, dof, expected = stats.chi2_contingency(contingency_table)

# Calculate Cramér's V
# Total number of observations
n = contingency_table.sum().sum()
# Minimum dimension of the contingency table minus 1
min_dim = min(contingency_table.shape) - 1
cramers_v = np.sqrt(chi2 / (n * min_dim))
print('Chi squared test:', chi2)
print('P-value for chi squared test:', p)
print('Cramers_v value:',cramers_v)
```

```
Chi squared test: 0.15902193516217972
P-value for chi squared test: 0.923567890607327
Cramers_v value: 0.003987755448396751
```

In [17]:
```
# Bivariate plot 2: Bar plot of Readmission by Complication Risk
sns.countplot(x='Complication_risk', hue='ReAdmis', data=df)
plt.title('Readmission by Complication Risk')
plt.xlabel('Complication Risk')
plt.ylabel('Count')
plt.show()
```


Readmission by Complication Risk
ReAdmis
No
Yes
Count
0
500
1000
1500
2000
2500
Medium
High
Low
Complication Risk

In [ ]: